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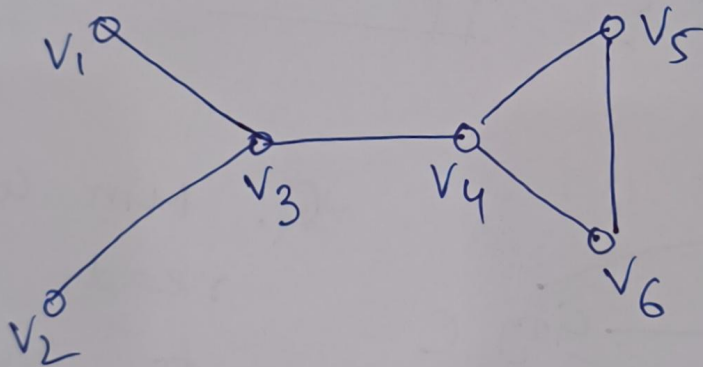
Graphs

A graph is a pair of sets (V, E)

V : Non-empty set of items.
called - vertices / nodes.

E : A (possibly empty) set of
2-item subset of V .
called edges.

Ex.

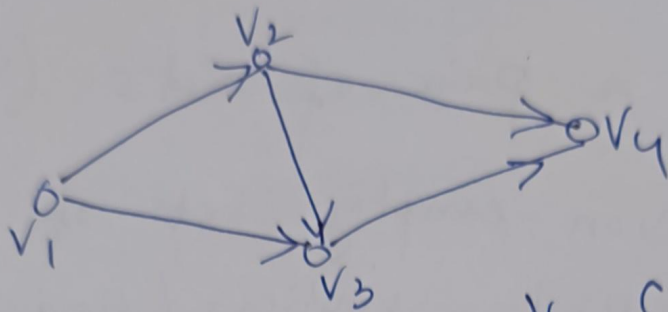


$$V = \{V_1, V_2, V_3, V_4, V_5\}$$

$$E = \{ \{V_1, V_3\}, \{V_2, V_3\}, \{V_3, V_4\}, \\ \{V_4, V_5\}, \{V_4, V_6\}, \{V_5, V_6\} \}$$

- Edges can be undirected (unordered)
pair
or directed (ordered pair)

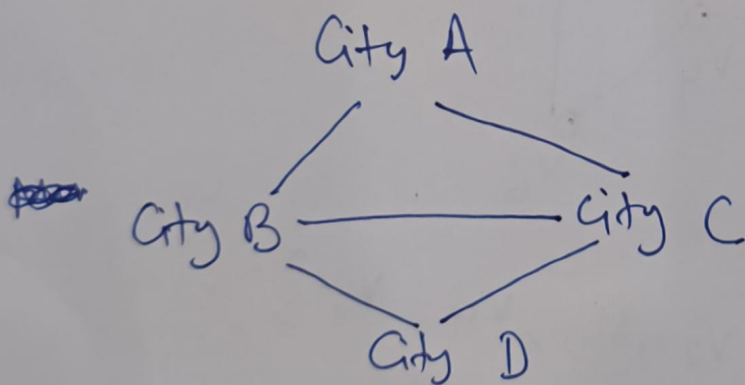
Directed graph Ex.



$$V = \{v_1, v_2, v_3, v_4\}$$

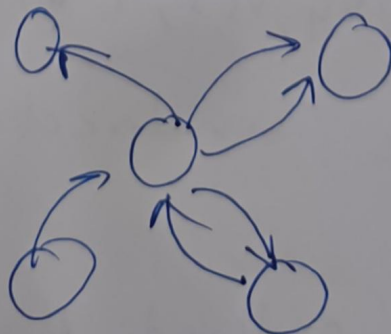
$$E = \{\{v_1, v_2\}, \{v_2, v_3\}, \{v_1, v_3\}, \{v_3, v_4\}, \{v_2, v_4\}\}$$

Why graphs? Applications



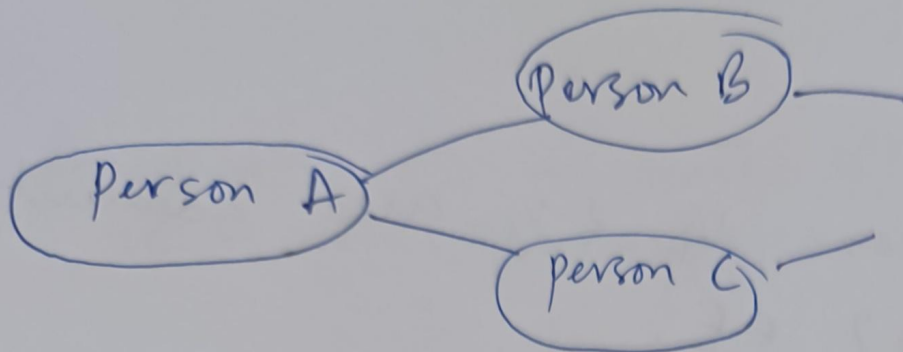
Road networks.

Q. Min cost to reach one city to another.



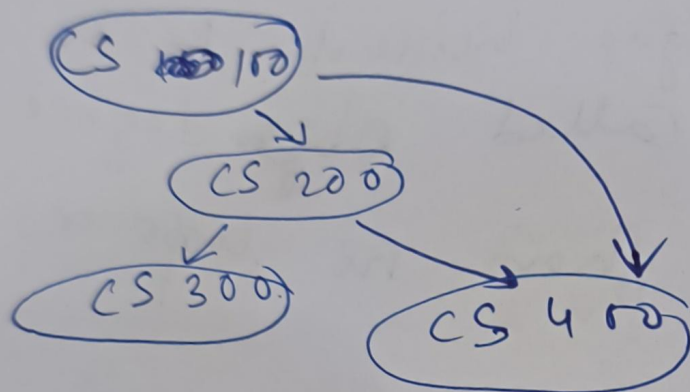
Web-related queries

Web graphs



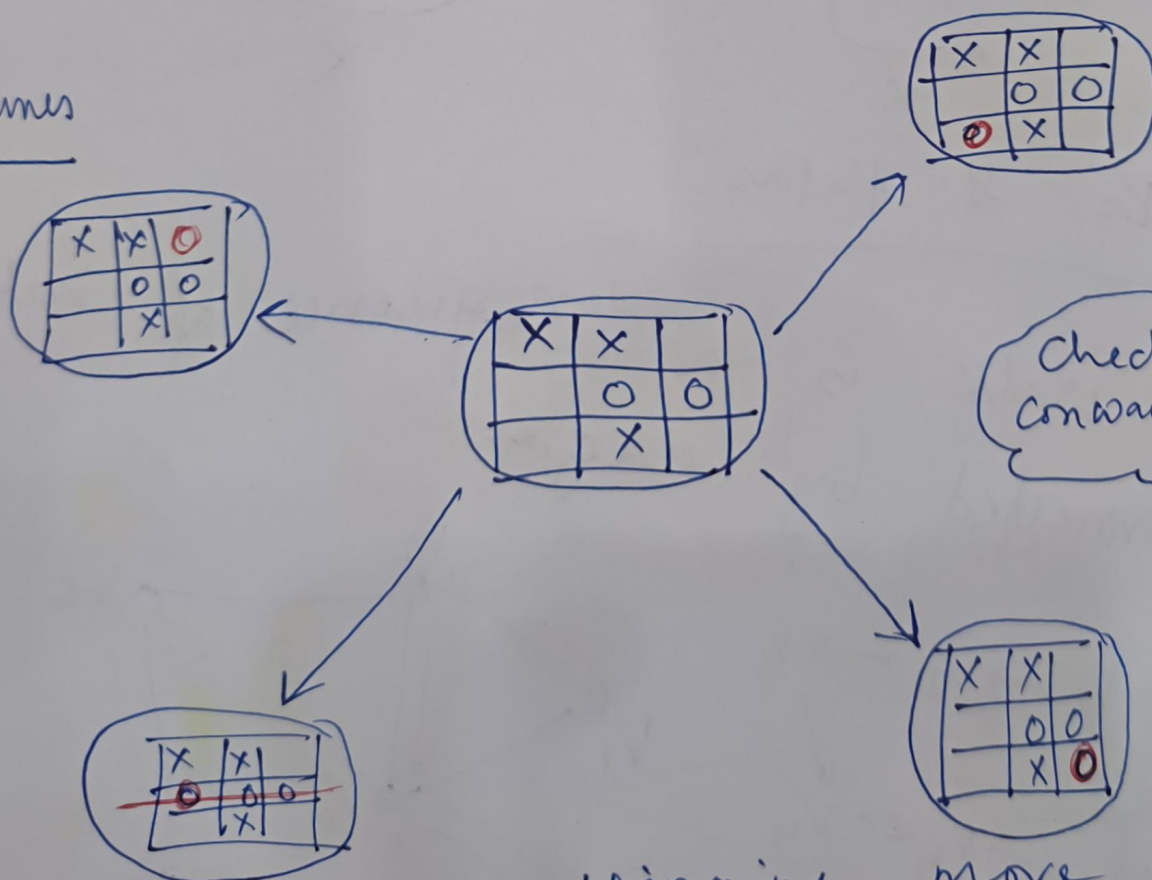
Social graphs

Q. Betⁿ two individuals
What's the # of hops.



precedence constraints.

Games



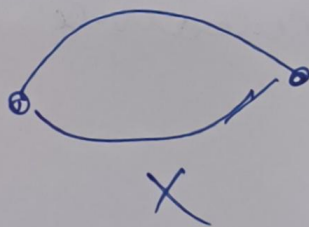
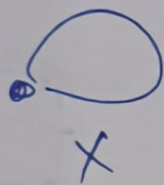
Check out
Conway's game
of life!

winning move -

Basic defⁿ.

- Two vertices v_i & v_j are adjacent if $\{v_i, v_j\} \in E$.
- An edge $e = \{v_i, v_j\}$ is incident to the vertices v_i & v_j .
- The # of edges incident to a vertex is called ~~edges~~ degree.

A simple graph has no loops or multi edges.

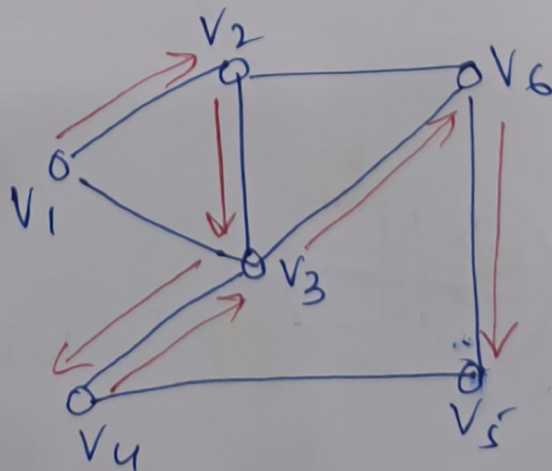


Walks & Paths

- A walk is a sequence of vertices connected by edges.

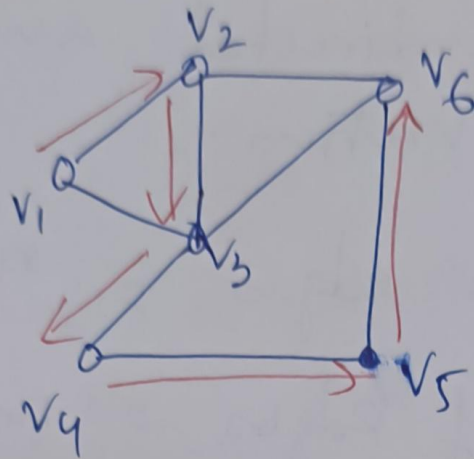
E.g. - $v_1 \rightarrow v_2 \rightarrow v_3$

$v_5 \leftarrow v_6 \leftarrow v_3 \leftarrow v_4$



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A path is a walk where all vertices are different.



$v_1 \rightarrow v_2 \rightarrow v_3 \rightarrow v_4$
 $\downarrow$
 $v_6 \leftarrow v_5$

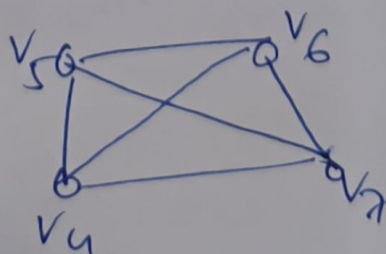
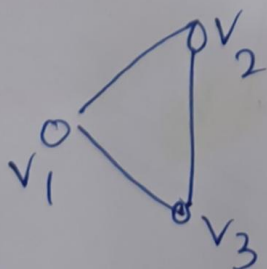
Lemma :- For any two distinct vertices u & v .

there exists a walk betⁿ u & v $\Leftrightarrow$ there exists a path betⁿ u & v .

proof :- ex.

Connectivity

Pair of vertices u & v are connected if there is a path betⁿ u & v .



v_1 & v_7 not connected.

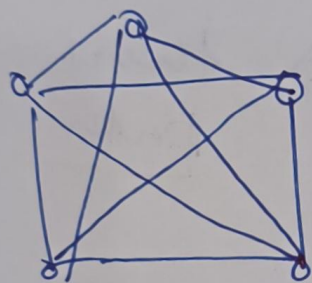
$v_5 - v_7$ is

A graph is connected if every pair of vertices are connected.

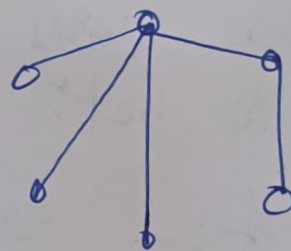
Given an undirected and connected graph of n vertices.

$$\text{max. \# of edges} = {}^nC_2$$

$$\text{min. \# of edges} = n-1$$



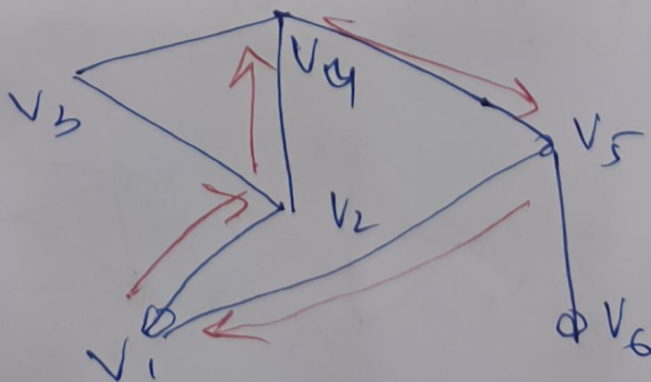
Complete graph.



tree

Cycles & Closed walk

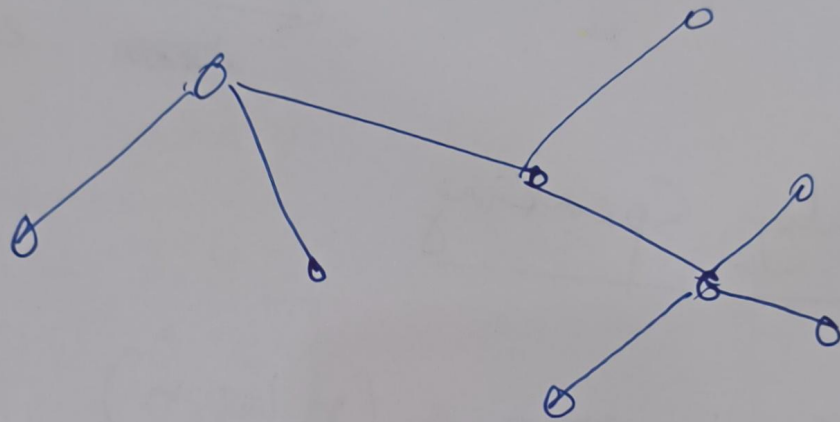
A closed walk is ~~also~~ a walk that starts and ends at the same vertex.



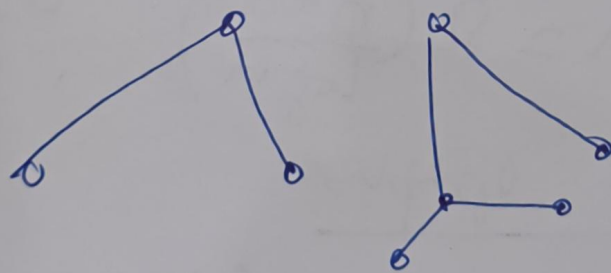
A cycle is a closed walk in which all vertices (except start & end, are diff.)

Trees

A tree is a connected and acyclic graph.



What about.



not a tree,
a forest.

Primary essence of tree (due to acyclic nature) is that trees are sparse.

Q. What are the other sparse graphs.

Sparse vs. Dense graphs

$n = \#$ of vertices.

$m = \#$ of edges.

Many applications. (Routing / Geosp. ind. / Robotics)
(but not always)

m is $\Omega(n)$ and $O(n^2)$.

$\longleftrightarrow$ ends of the spectrum.

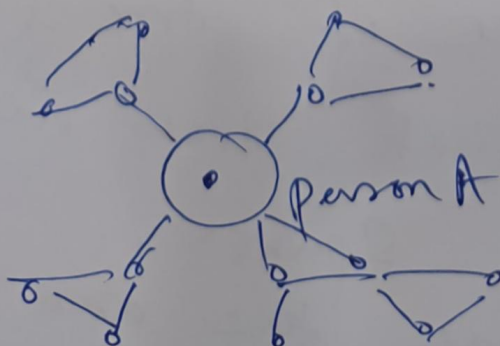
Roughly Speaking

Sparse: $m = O(n \log n)$ close to linear.

Dense: $m = \Omega\left(\frac{n^2}{\log n}\right)$ close to quadratic.

Exs. of Sparse graphs

Social media graphs (fb. / linkedin)

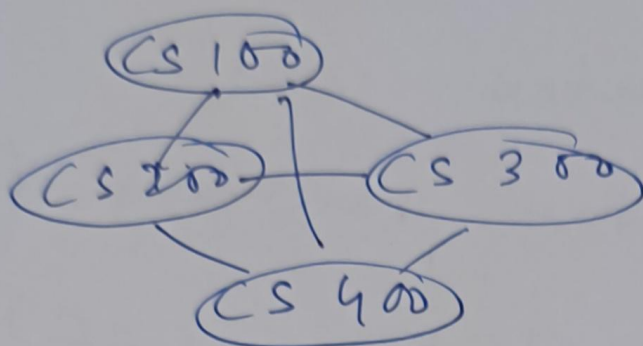


(Sometimes nicely clusterable!).

But could be also dense.

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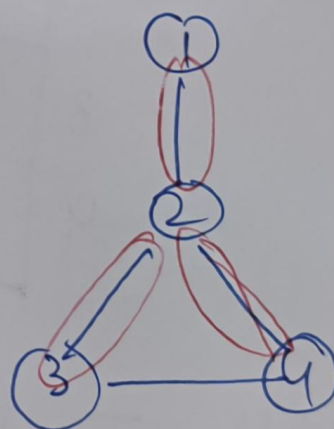
For instance ~~prefer~~ pref. graphs.



Representing graphs

Adjacency matrix.

$$A = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$



Symmetric

$\Theta(n^2)$ - Space.

This is OK for dense graphs.

But not OK for sparse graphs.

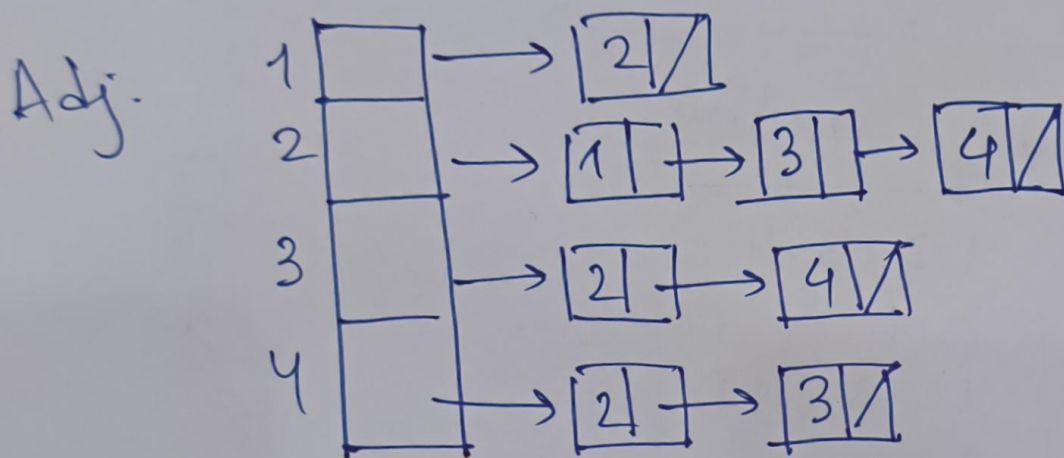
other options. (Adjacency list)

Array of $|V|$ linked list.

for each vertex $u \in V$.

Adj. [u] Stores neighbors of u

$$\hookrightarrow \{v \in V : \{u, v\} \in E\}$$



$$\text{Space} = \Theta(|V| + |E|) \quad \text{or} \quad \Theta(m+n)$$

Which is better?

— Depends on the applications & density of the i/g graph.

Adjacency list

- Great for "follow your nose" operation.
- Better for massive sparse graphs. (e.g. Web graph).